# Abhinav Meethele Valappil

Backend Engineer · TypeScript · NestJS · ClickHouse · Web3

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## SUMMARY

Backend engineer with 5+ years of experience, including four years at Moralis building blockchain data APIs and real-time indexing pipelines for Solana and EVM chains. Experienced in TypeScript and NestJS with PostgreSQL, DynamoDB, ClickHouse and event streaming, and in Rust for performance-critical services. Comfortable owning services from design through production, with AI-assisted development as part of the daily workflow.

## EXPERIENCE

**Software Developer** · Moralis Web3 · Remote — Sep 2022 – Present

*Backend engineer on the team behind Moralis' Web3 data APIs, which serve wallet, token and price data for Solana and EVM chains.*

- **Designed and built a Solana DEX indexer from the ground up**, with a modular parser framework covering 10 pool types across Orca, Raydium, Meteora and PumpSwap, plus the pair, swap, reserve and candlestick APIs built on top of it.
- Developed real-time streaming pipelines that turn raw on-chain data into token prices, metadata and OHLCV candles, and extended swap indexing to Uniswap v3/v4, PancakeSwap Infinity and Algebra-based DEXes on EVM chains.
- Reduced API response times for large Solana wallets from 7–8 seconds to around 2.5 seconds by parallelising metadata lookups, resolving timeouts for users holding thousands of tokens.
- Optimised ClickHouse queries that were scanning hundreds of millions of rows, consolidating 87 queries into 4 that complete in about 2 seconds.
- Ran a storage evaluation for a token-filtering API, benchmarking Elasticsearch against ClickHouse on 1M records, and chose Elasticsearch for 3–6x faster reads and twice the write throughput.
- Migrated high-traffic wallet, transfer and swap APIs to ClickHouse using shadow testing, comparing old and new responses in production before each cutover.
- Lowered peak memory usage of a Rust data-correction service by up to 19% with no change in output.
- Worked with API customers to investigate reported bugs and data gaps, and delivered fixes to the endpoints they relied on.
- Managed Kubernetes deployments with Helm, load-tested APIs with k6, and built service monitoring in Grafana, Prometheus and Datadog.
- Use Claude Code daily for design, code review and debugging, validating its output against tests and production metrics. Built an internal AI agent that detects API latency regressions and suggests likely root causes.

**System Engineer** · Tata Consultancy Services · Delhi — Jan 2021 – Aug 2022

- Developed and maintained Oracle Siebel CRM workflows for SBI Card's credit card customer support platform.
- Started as an Assistant System Engineer (apprentice) in Jan 2021 and was promoted to System Engineer in Dec 2021.

**Data Entry Associate** · Awign · Remote — Jan 2019 – Jan 2021

- Handled remote data transcription for Zomato.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | TypeScript, JavaScript, Python, Rust, SQL, C, Java |
| **Backend** | Node.js, NestJS, REST, gRPC/Protobuf, Kafka-compatible streaming, BullMQ, Redis |
| **Databases** | PostgreSQL, DynamoDB, MongoDB, ClickHouse, Elasticsearch, OpenSearch, MySQL |
| **Cloud & infra** | AWS, Docker, Kubernetes, Helm, GitLab CI, k6 |
| **Observability** | Grafana, Prometheus, Datadog, Slack alerting |
| **Web3** | Solana and EVM chains, ethers.js, @solana/web3.js, smart-contract event decoding, DEX protocols |
| **AI** | Claude Code, Claude Agent SDK, Anthropic and OpenAI APIs |
| **Other** | React, Next.js, Oracle Siebel CRM |

## EDUCATION

**B.Tech, Computer Science & Engineering** — 2016 – 2020 · CGPA 8.46/10
Government College of Engineering Kannur

**Class 12 (CBSE)** — 2016 · 85.4%
St. Francis School Indirapuram

**Class 10 (CBSE)** — 2014 · CGPA 8.8/10
St. Francis School Indirapuram

## ACADEMIC PROJECTS

- **Fake News Detection** (final year): news classifier built with LightGBM and WordBatch.
- **Online Bookstore**: full-stack e-commerce app built with Node.js and MySQL.
- **Spam Filter**: text classifier built with Python and NLTK.

## CERTIFICATIONS & ACHIEVEMENTS

- LinkedIn Skill Assessments: Python, React.js
- NPTEL Intro to Machine Learning (IIT Kharagpur), 80% · 2019
- Android app development training and certification · 2018
- 1st place, ROBOCEK robotics event, GCE Kannur · 2016